

Master Thesis Proposal

Study Programme: MSc Computer Science – Big Data and AI

Name Surname: Sultanov Suleiman Abdutalipovich

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Master Thesis Topic: Adaptive Hybrid Kernel SVMs Dynamic Feature Weighting for Noisy, High-Dimensional IoT Healthcare Data

Structure:

Name of First Supervisor: Dr. Alexander I. Iliev

Name of Professorial Supervisor:
(In case first Supervisor is a scientific worker)

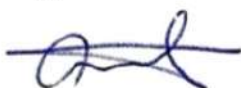
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Date, Candidate's Signature

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Prof Dr Alexander I. Iliev

Date, SRH Supervisor's Signature

Date, SRH/External Supervisor's Signature
(In case first Supervisor is a scientific worker)

Adaptive Hybrid Kernel SVMs

Dynamic Feature Weighting for Noisy, High-Dimensional IoT Healthcare Data

1. Introduction

1.1 Background and motivation

The Internet of Medical Things (IoMT) and wearable sensing now generate continuous, high-dimensional data streams from multiple modalities. Modern healthcare systems rely on sensor networks that track physiological signals such as heart rate, blood oxygen, ECG waveforms, accelerometer-based motion, and ambient conditions. These signals are usually noisy and nonstationary. Sensors vary, the environment interferes with them, and patients differ physiologically from one another, which makes the resulting data difficult for classical machine learning methods that assume clean, stationary inputs (Al-amri et al., 2021; AL-Dhief et al., 2020).

To use this data for clinical decision support, a classifier has to (i) handle high dimensionality without overfitting, (ii) stay robust to the noise and outliers that sensor-based collection inevitably produces, and (iii) adapt to shifting data characteristics without being re-tuned or retrained from scratch every time. Elderly care is a particularly hard case. Older adults often wear several monitoring devices at once, adherence is inconsistent, and the clinical relevance of a given feature can change as health conditions evolve (Talaat & El-Balka, 2023; Abdellatif et al., 2022).

Support Vector Machines (SVMs) are a reasonable starting point for high-dimensional classification and regression. They have a solid grounding in statistical learning theory, generalization guarantees via margin maximization, and flexible nonlinear modeling through the kernel trick (Vallabhuni & Debasis, 2025; Wang et al., 2023; Muriira et al., 2018). The kernel trick implicitly maps data into a higher-dimensional space where linear separation becomes feasible, which is useful for biomedical classification problems with nonlinear decision boundaries.

The trouble is that standard SVMs with fixed kernels tend to struggle on noisy, multimodal healthcare data from IoT devices. A single kernel function rarely captures the diverse structure present in heterogeneous sensor data. A linear kernel picks up broad trends but misses local nonlinearities. An RBF kernel models local structure but can overfit to noise in high-dimensional spaces. On top of that, the relevance and reliability of individual features can change over time and across patients, so a static kernel choice is not well suited to dynamic IoT healthcare environments (Al-hajjar & Al-Qurabat, 2023; Mahdavinejad et al., 2018).

1.2 Research gap and novelty statement

Multiple kernel learning (MKL) methods have improved on single-kernel SVMs by learning a convex combination of base kernels, so that different kernel functions can capture different aspects of the data

(Gönen & Alpaydin, 2011; Rakotomamonjy et al., 2008). Separately, weighted SVM variants have been proposed to handle noisy or imbalanced features by assigning different importance to different feature dimensions. These two lines of research have largely developed independently, and their integration in the IoT healthcare setting has received little attention.

No existing approach, as far as we are aware, handles the following combination jointly: (1) data-quality-driven, per-feature weight adaptation built directly into the kernel evaluation, where feature weights respond to measurable indicators of sensor reliability such as missingness rates and variance stability; (2) adaptive fusion of global and local kernel components that balances the complementary strengths of linear and RBF kernels; and (3) targeted application to noisy, high-dimensional IoT healthcare streams, where these problems are most acute.

Standard MKL approaches (Gönen & Alpaydin, 2011; Rakotomamonjy et al., 2008) learn fixed kernel combination weights during training and do not update them at inference time. Feature-weighted SVMs, for example the feature-and-sample weighted SVM (Zhang et al., 2011) and the weighted SVM of Yang et al. (2005), typically use static feature weights derived from preprocessing steps such as mutual information or variance thresholds, without coupling them to the kernel composition. Fuzzy SVM approaches address noise sensitivity by assigning membership degrees to samples (Lin & Wang, 2004), but they do not feed per-feature data-quality indicators into the kernel evaluation itself. Our contribution bridges these lines of work: feature weights modulate the kernel evaluation directly, so unreliable features are de-emphasized inside the similarity computation rather than filtered out beforehand.

1.3 Scope of this thesis

To keep the project feasible within a Master's thesis timeline, this work focuses on a single core contribution: designing and evaluating an adaptive hybrid kernel SVM that combines dynamic feature weighting with an adaptive linear-RBF kernel mixture. The scope is deliberately narrowed so the core idea can be developed and evaluated carefully. The thesis will:

- Formulate a composite kernel that combines a global linear component and a local RBF component with a tunable mixture coefficient, where both components operate on feature-weighted inputs.
- Develop a feature weighting mechanism that computes per-feature reliability scores from missingness rates and variance stability, and feeds these weights directly into the kernel computation.
- Provide a complete mathematical formulation, including the optimization objective, feature weight computation, and algorithmic pseudocode.
- Evaluate the framework on two public benchmark datasets (OPPORTUNITY and CASAS) against a focused set of baselines, using accuracy, F1-score, and noise robustness as the primary criteria.

Out of scope: the following extensions are deferred to future work so the project stays feasible: online or streaming adaptation of kernel weights at inference, edge device deployment and optimization, fairness auditing across demographic subgroups, privacy-preserving computation (for example federated learning or differential privacy), and deep learning hybrid architectures. Section 8 returns to these as future directions.

2. Literature Review

Three areas are directly relevant to the proposed contribution: kernel methods for biomedical data classification, multiple kernel learning, and noise handling for IoT healthcare data. The review below draws on peer-reviewed work from IEEE, ACM, Elsevier, Springer, and MDPI venues, and its purpose is to locate the specific gap this thesis addresses.

2.1 SVMs and kernel methods for high-dimensional biomedical data

Since their formalization in the 1990s, Support Vector Machines have been widely used for classification in high-dimensional spaces. The central idea is to construct a maximum-margin hyperplane that separates classes with the largest possible gap, which yields generalization guarantees through Vapnik-Chervonenkis theory. The kernel trick extends this to nonlinear classification by implicitly mapping input features into a higher-dimensional space where linear separation becomes feasible.

SVMs have been widely adopted for biomedical data because they work well in the high-dimensional, small-sample-size regime typical of clinical settings. Vallabhuni and Debasis (2025) show that kernel-based approaches are still relevant for IoT-based health monitoring when paired with feature extraction tailored to physiological signals. Wang et al. (2023) review biomedical prediction systems and note that kernel methods remain competitive with deep learning when training data is limited or when interpretability matters.

The choice of kernel has a large effect on SVM performance. The most common options are the linear kernel $K(x,y) = x^T y$, which works when data is roughly linearly separable; the polynomial kernel, which captures feature interactions up to a specified degree; and the Radial Basis Function (RBF) kernel $K(x,y) = \exp(-\gamma \|x-y\|^2)$, which measures similarity from Euclidean distance and can model highly nonlinear decision boundaries. The RBF kernel is popular in biomedical applications because it is flexible, but its performance depends strongly on the bandwidth parameter γ and it can overfit in high-dimensional spaces with noisy features (Al-hajjar & Al-Qurabat, 2023; Muriira et al., 2018).

A limitation of single-kernel approaches is that they cannot capture both global trends and local nonlinear structure at the same time in heterogeneous data. A linear kernel captures broad relationships across features but cannot model nonlinear interactions. An RBF kernel with a small bandwidth captures local structure but can miss global patterns. For multimodal IoT healthcare data, where different sensor modalities have fundamentally different characteristics, no single kernel is likely to be adequate.

2.2 Multiple kernel learning (MKL)

Multiple Kernel Learning addresses the single-kernel limitation by learning an optimal combination of several base kernels. The general form is a weighted sum $K = \sum \beta_k K_k$, where the weights β_k are non-negative and sum to one, and K_k are the individual base kernels. The weights are learned jointly with the SVM parameters, so the model can determine how much to rely on each kernel automatically.

Gönen and Alpaydin (2011) survey MKL algorithms and group them by optimization approach (fixed rules, heuristic, optimization-based), the form of kernel combination (linear, nonlinear, data-dependent), and the learning methodology (one-step, two-step, alternating). They report that MKL tends to outperform single-kernel methods when the data has heterogeneous structure that different kernels pick up differently. Rakotomamonjy et al. (2008) propose SimpleMKL, a gradient descent method on kernel weights with an L1-norm constraint that is competitive while being cheaper to train.

MKL has produced useful results in biomedical and IoT contexts. Wang et al. (2023) and Al-amri et al. (2021) report that combining kernels with different properties (for example a linear kernel for overall trends and an RBF kernel for local nonlinearities) gives classification gains over single-kernel counterparts in healthcare analytics. Muriira et al. (2018) show that kernel combination strategies are effective for wireless sensor network data, a setting with many of the same characteristics as IoT healthcare.

Standard MKL has one important limitation for streaming IoT data: the kernel combination weights are learned during training and stay fixed at inference time (Gönen & Alpaydin, 2011; Sonnenburg et al., 2006). If data quality changes after training, say a sensor becomes unreliable or environmental conditions shift, the static kernel weights cannot adapt. Standard MKL also operates at the kernel level and ignores per-feature differences in data quality. A noisy feature contributes to every kernel computation equally regardless of its reliability. These limitations are the main motivation for integrating data-quality-driven feature weighting into the MKL framework.

2.3 Noise handling and feature weighting in IoT healthcare

Noise, sensor drift, and missingness are inherent to IoT healthcare data. Al-amri et al. (2021) list noise handling as one of the central challenges for machine learning in IoT environments, pointing to calibration errors, wireless transmission artifacts, and environmental interference. AL-Dhief et al. (2020) describe similar problems in voice pathology surveillance, where signal quality varies a lot across recording conditions. Talaat and El-Balka (2023) document noise in wearable stress monitoring, where motion artifacts and electrode displacement introduce systematic distortions.

Several strategies have been proposed to deal with this. At the preprocessing level, robust normalization, outlier detection, and missing value imputation can reduce the impact of noisy measurements before they reach the classifier. Abdellatif et al. (2022) argue that preprocessing pipelines should account for the specific noise characteristics of different sensor modalities. Mahdavinejad et al. (2018) survey machine learning for IoT data analysis and argue for learning algorithms that are inherently robust to noise, rather than relying entirely on preprocessing.

Feature weighting is a more principled approach to noise handling: each feature dimension gets a weight that reflects its informativeness or reliability. Traditional methods derive these weights from statistical measures like mutual information, Fisher's discriminant ratio, or variance thresholds (Zhang et al., 2011; Yang et al., 2005). These methods usually compute weights once during preprocessing and apply them statically, with no account for temporal changes in reliability that are common in IoT deployments.

Feature weights computed independently of the kernel may also interact poorly with the kernel's similarity measure (Lin & Wang, 2004).

Integrating feature weighting directly into the kernel evaluation, so that reliability modulates the similarity computation itself, has been underexplored. This is the direction that matters for IoT healthcare, where sensor reliability can shift over time and different features may need different amounts of emphasis depending on current data quality. The proposed framework targets this gap: feature weights are computed from data-quality indicators and applied inside the kernel function, so unreliable features contribute less to the similarity measure used for classification.

2.4 Summary of gaps

The literature supports both the value of hybrid kernels for heterogeneous biomedical data and the importance of feature weighting for noisy IoT healthcare streams. Three specific gaps come out of the review:

1. Existing MKL approaches learn static kernel combination weights that cannot adapt to changing data quality at inference time.
2. Feature weighting methods typically operate as preprocessing steps decoupled from the kernel learning objective, which can produce suboptimal interaction between the weights and the kernel similarity.
3. No prior work jointly optimizes adaptive kernel mixture coefficients and data-quality-driven feature weights within a single SVM framework designed for IoT healthcare data.

Together, these gaps motivate the proposed Adaptive Hybrid Kernel SVM, which integrates feature weighting directly into a composite linear-RBF kernel to address the specific problems of noisy, high-dimensional IoT healthcare data.

3. Research Objectives and Hypotheses

3.1 Primary objective

The primary objective is to design and evaluate an adaptive hybrid kernel SVM that combines a global linear kernel and a local RBF kernel with (a) an adaptive kernel mixture coefficient and (b) data-quality-driven per-feature weights, for classification of noisy, high-dimensional IoT healthcare data. The framework should show measurable gains over standard single-kernel SVMs and static MKL baselines on classification accuracy, F1-score, and noise robustness.

3.2 Sub-objectives

4. Mathematical formulation: derive a precise formulation of the adaptive hybrid kernel, including the composite kernel function, the feature weight computation from data-quality indicators, the SVM optimization objective, and the hyperparameter search space.
5. Implementation: implement the framework as an extension of standard SVM solvers in Python and scikit-learn, using a custom precomputed kernel matrix that incorporates both the adaptive kernel mixture and the feature weighting.
6. Empirical validation: evaluate the method on two public benchmark datasets (OPPORTUNITY for multimodal sensor fusion and CASAS for ambient elderly monitoring) against three baselines: a standard SVM with RBF kernel, a standard SVM with linear kernel, and a simple static MKL approach.
7. Noise robustness analysis: evaluate robustness to synthetic noise injection at varying intensity levels, to quantify the benefit of data-quality-driven feature weighting.

3.3 Hypotheses

Hypothesis H1 (hybrid kernel advantage): on multimodal, noisy IoT healthcare datasets, the adaptive hybrid kernel SVM (combining linear and RBF kernels with feature weighting) will achieve higher classification accuracy and macro F1-score than any single fixed kernel (RBF or linear alone). The MKL literature suggests that combining complementary kernels captures heterogeneous data structure more effectively than any single kernel (Gönen & Alpaydin, 2011; Wang et al., 2023; Muriira et al., 2018).

Hypothesis H2 (noise resilience): incorporating data-quality-driven feature weights into the kernel evaluation will give better robustness to synthetic noise injection than standard kernel SVMs or a static MKL baseline without feature weighting. In concrete terms, the proposed method should show smaller performance degradation as noise intensity increases. Prior findings on noise-aware learning strategies in high-dimensional settings support this direction (Al-amri et al., 2021; Mahdavinnejad et al., 2018; Abdellatif et al., 2022).

Hypothesis H3 (feature weighting contribution): the data-quality-driven feature weighting mechanism will produce a measurable improvement over the hybrid kernel alone (that is, the same linear-RBF

mixture without feature weighting), isolating the contribution of adaptive feature weighting. Earlier work showing that feature weighting improves noise immunity in SVM classification (Zhang et al., 2011; Yang et al., 2005), together with the standard use of ablation studies for isolating MKL components (Gönen & Alpaydin, 2011), motivates this hypothesis. H3 is tested through an ablation comparing the full model against a variant without feature weighting.

4. Methodology

What follows is the methodological framework for the Adaptive Hybrid Kernel SVM: the mathematical formulation, algorithm design, data handling, experimental protocol, and implementation details. The design aims to be reproducible and feasible within a Master's thesis.

4.1 Proposed model: Adaptive Hybrid Kernel SVM

The core of the proposed framework is a composite kernel that combines two complementary kernel functions with data-quality-driven feature weighting. The design rationale is straightforward: the linear kernel component captures broad, cross-feature structure and global trends, while the RBF kernel component handles fine-grained nonlinear interactions and local similarity. Feature weighting makes sure that unreliable or noisy features contribute less to both components, so the similarity measure itself is more robust.

Let $x \in \mathbb{R}^d$ denote a d -dimensional input sample. The composite kernel K is defined as:

$$K(x_i, x_j) = \mu \cdot K_{\text{lin}}(W \odot x_i, W \odot x_j) + (1 - \mu) \cdot K_{\text{rbf}}(W \odot x_i, W \odot x_j)$$

The components are:

- Feature weight vector W : $W = (w_1, w_2, \dots, w_d)$ is a vector of per-feature weights, each $w_k \in [0, 1]$ reflecting the estimated reliability of feature k . The operation $W \odot x$ is element-wise multiplication (Hadamard product), which scales each feature by its reliability weight before the kernel computation.
- Kernel mixture coefficient μ : $\mu \in [0, 1]$ controls the balance between the global linear component and the local RBF component. At $\mu = 1$, the model reduces to a feature-weighted linear SVM; at $\mu = 0$, it reduces to a feature-weighted RBF SVM. Intermediate values blend both.
- Linear kernel: $K_{\text{lin}}(a, b) = a^T b$ is the inner product between feature-weighted inputs, capturing global linear relationships across the feature space.
- RBF kernel: $K_{\text{rbf}}(a, b) = \exp(-\gamma \|a - b\|^2)$ measures local similarity from the Euclidean distance between feature-weighted inputs, with bandwidth parameter γ controlling the scale of locality.

Because the feature weights are applied before the kernel computation, unreliable features are de-emphasized within the similarity measure itself. This is different from feature selection or weighting applied as a separate preprocessing step, because the kernel function's behavior, including how it measures distances and inner products, is directly modulated by feature reliability. A feature with weight $w_k \approx 0$ is effectively invisible to both kernel components, while a fully reliable feature with $w_k = 1$ contributes at its natural scale.

4.2 Feature weight computation

The feature weights w_k are derived from two complementary data-quality indicators that capture different aspects of reliability. Both can be computed from the training data alone without external metadata, and they target the most common failure modes of IoT sensors: data loss (missingness) and measurement instability (variance fluctuation).

Missingness rate. For each feature k , we compute the fraction of training samples for which the feature value is missing or invalid: $m_k = (\text{number of missing or invalid values in feature } k) / (\text{total number of training samples})$. Features with high missingness are systematically unreliable and should get lower weights. This captures sensor dropout and communication failures, which are common in IoT deployments.

Variance stability. For each feature k , we compute the coefficient of variation (CV): $s_k = \sigma_k / |\mu_k|$, where σ_k is the standard deviation and μ_k is the mean of feature k over the training data. A high CV means the feature's values are highly dispersed relative to its mean, which can indicate sensor noise, calibration drift, or environmental interference. For features with near-zero mean, we add a small constant ϵ to the denominator to avoid division by zero.

The combined feature weight is:

$$w_k = (1 - m_k) \cdot \sigma(-\lambda \cdot s_k)$$

where $\sigma(z) = 1/(1 + \exp(-z))$ is the sigmoid function and $\lambda > 0$ is a sensitivity hyperparameter that controls how aggressively features with high variance instability are penalized. The sigmoid maps variance stability to a smooth weight between 0 and 1: features with low variance instability get weights near 1, and features with high instability get weights closer to 0.5 (or lower for very high instability).

The multiplicative structure $(1 - m_k) \cdot \sigma(-\lambda \cdot s_k)$ keeps the two indicators independent: a feature with 80% missing values gets a maximum weight of 0.2 no matter how stable its variance is, while a fully observed feature with high variance instability is penalized only through the sigmoid term. After computing all feature weights, we normalize them so that $\sum_k w_k = d$. This preserves the overall scale of the feature space and keeps the regularization parameter C and the RBF bandwidth γ interpretable relative to standard SVMs.

4.3 Kernel mixture coefficient optimization

The mixture coefficient μ is treated as a hyperparameter and optimized jointly with the SVM regularization parameter C and the RBF bandwidth γ via grid search with cross-validation. This gives a three-dimensional hyperparameter search over:

- $\mu \in \{0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0\}$
- $C \in \{0.01, 0.1, 1, 10, 100\}$
- $\gamma \in \{0.001, 0.01, 0.1, 1, 10\}$

The feature weight sensitivity parameter λ is searched over $\{0.1, 0.5, 1.0, 2.0, 5.0\}$. The total grid is $11 \times 5 \times 5 = 1,375$ combinations, which is feasible with 5-fold cross-validation at the dataset sizes we are working with. The combination that maximizes mean cross-validation accuracy on the training set is selected for final evaluation on the test set.

4.4 Optimization objective

Given the composite kernel matrix K computed from feature-weighted inputs, the SVM optimization follows the standard dual formulation. Let $\{(x_1, y_1), \dots, (x_n, y_n)\}$ be the training set with labels $y_i \in \{-1, +1\}$. The dual objective is:

$$\begin{aligned} \text{maximize } L(\alpha) &= \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j K(x_i, x_j) \\ \text{subject to: } &0 \leq \alpha_i \leq C \text{ for all } i, \text{ and } \sum_i \alpha_i y_i = 0 \end{aligned}$$

The decision function for a new sample x is:

$$f(x) = \text{sign}(\sum_i \alpha_i y_i K(x_i, x) + b)$$

where b is the bias term computed from the support vectors. A key design choice is that the feature weights W are computed before kernel matrix construction and are not optimized jointly with the dual variables α . This keeps the kernel matrix K positive semi-definite (it is a non-negative weighted sum of valid kernel matrices), and the optimization remains a standard convex quadratic program solvable by any SVM solver (for example LIBSVM via scikit-learn's SVC with a precomputed kernel). Separating the two also keeps the computational cost manageable and avoids the non-convex optimization that would come from joint optimization of feature weights and SVM parameters.

4.5 Algorithm (pseudocode)

Algorithm 1: Adaptive Hybrid Kernel SVM — Training

Input: Training data $X \in \mathbb{R}^{n \times d}$, labels $y \in \{-1, +1\}^n$

Hyperparameters: μ, C, γ, λ

Output: Trained SVM model (support vectors, α^* , bias b^*)

Step 1: COMPUTE DATA-QUALITY INDICATORS

```
for each feature  $k = 1, \dots, d$ :
     $m_k \leftarrow \text{count\_missing}(X[:, k]) / n$ 
     $s_k \leftarrow \text{std}(X[:, k]) / (|\text{mean}(X[:, k])| + \epsilon)$ 
```

Step 2: COMPUTE FEATURE WEIGHTS

```
for each feature  $k = 1, \dots, d$ :
     $w_k \leftarrow (1 - m_k) \cdot \text{sigmoid}(-\lambda \cdot s_k)$ 
Normalize:  $w \leftarrow w \cdot (d / \sum_k w_k)$ 
```

Step 3: APPLY FEATURE SCALING

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 $\tilde{X} \leftarrow X \odot W$  (element-wise multiplication)
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Step 4: CONSTRUCT COMPOSITE KERNEL MATRIX

$$K_{lin} \leftarrow \tilde{X} \cdot \tilde{X}^T$$

$$K_{rbf}[i,j] \leftarrow \exp(-\gamma \cdot \|\tilde{X}[i] - \tilde{X}[j]\|^2) \text{ for all } i,j$$

$$K \leftarrow \mu \cdot K_{lin} + (1 - \mu) \cdot K_{rbf}$$

Step 5: SOLVE SVM DUAL PROBLEM
 $(\alpha^*, b^*) \leftarrow \text{SVM_Dual_Solver}(K, y, C)$

Step 6: RETURN (support_vectors, α^* , b^* , W , μ , γ)

Algorithm 2: Adaptive Hybrid Kernel SVM — Prediction

Input: New sample x_{new} , trained model (SVs, α^* , b^* , W , μ , γ)
Output: Predicted label y_{pred}

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1. $\tilde{x}_{new} \leftarrow W \odot x_{new}$
 2. for each support vector $\tilde{x}_i$:
 $k_i \leftarrow \mu \cdot (\tilde{x}_i^T \tilde{x}_{new}) + (1-\mu) \cdot \exp(-\gamma \|\tilde{x}_i - \tilde{x}_{new}\|^2)$
 3. $y_{pred} \leftarrow \text{sign}(\sum_i \alpha^*_i \cdot y_i \cdot k_i + b^*)$

4.6 Datasets

The evaluation uses two public benchmark datasets chosen to cover complementary IoT healthcare scenarios. Two datasets balance breadth of evaluation against the feasibility constraints of a Master’s thesis.

Dataset	Type	Description	Relevance
OPPORTUNITY	Body-worn + ambient sensors	Multimodal sensor data for activity recognition; 113 features from accelerometers, gyroscopes, and ambient sensors; includes natural sensor dropout.	Stress-tests high-dimensional multimodal fusion under realistic sensor noise and dropout.
CASAS Smart Home	Ambient sensors	Motion sensors, door sensors, and temperature sensors for elderly activity monitoring in real smart home environments.	Evaluates ambient elderly monitoring with heterogeneous sensor types and temporal activity patterns.

Both datasets are public, widely used in IoT and ubiquitous computing research, and enable direct comparison with prior work. OPPORTUNITY (Chavarriaga et al., 2013) provides a high-dimensional setting with natural sensor failures and was originally published in Pattern Recognition Letters (Elsevier). The CASAS smart home project (Cook et al., 2013) targets elderly monitoring and was published in IEEE

Computer. Both have been used extensively for benchmarking activity recognition algorithms across IEEE, ACM, and Springer venues.

4.7 Data preprocessing

The same preprocessing pipeline is applied to both datasets to make comparisons fair:

8. Missing value handling. Missing values are imputed using forward-fill (last observation carried forward), followed by mean imputation for any remaining gaps. The missingness rate for each feature is recorded before imputation for use in the feature weight computation.
9. Robust normalization. Features are normalized using robust scaling (median subtraction and interquartile range division), which reduces the effect of outliers while preserving distributional information.
10. Feature extraction. For OPPORTUNITY, statistical features (mean, standard deviation, min, max, energy) are computed over sliding windows. For CASAS, sensor event counts and temporal features are extracted per activity window.
11. Train-test split. Data is split into training (80%) and test (20%) sets, stratified by class label. Cross-validation is performed only on the training set; the test set is reserved for final evaluation.

4.8 Baseline models

The proposed method is benchmarked against three baselines chosen to isolate the specific contributions of the hybrid kernel and the feature weighting:

Baseline	Description	Purpose
SVM-RBF	Standard SVM with RBF kernel, hyperparameters (C , γ) optimized via cross-validation. No feature weighting.	Tests whether the hybrid kernel and feature weighting improve over the most common single-kernel choice.
SVM-Linear	Standard SVM with linear kernel, regularization C optimized via cross-validation. No feature weighting.	Tests whether the nonlinear (RBF) component adds value beyond the linear baseline.
Static MKL	Convex combination of linear and RBF kernels with mixture weight μ optimized via cross-validation, but without feature weighting.	Isolates the contribution of feature weighting by comparing the hybrid kernel with and without data-quality weights.

4.9 Evaluation protocol

Cross-validation. 5-fold stratified cross-validation is used for hyperparameter selection on the training set. Stratification preserves class distributions in each fold. To assess stability, the entire experiment

(random split + cross-validation + test evaluation) is repeated 5 times with different random seeds, and results are reported as mean $\pm$ standard deviation.

Primary metrics. Classification accuracy and macro-averaged F1-score. Accuracy gives an overall measure of correctness, and macro F1-score accounts for class imbalance by averaging F1 across classes with equal weight.

Noise robustness test. Gaussian noise with standard deviation $\sigma \in \{0.1, 0.3, 0.5, 0.7, 1.0\}$ is injected into a randomly selected 30% of feature dimensions at test time. Performance degradation curves (accuracy and F1 versus noise level) are plotted for each method. This simulates real-world scenarios where some sensors degrade or fail during deployment.

Ablation study. To isolate the contribution of feature weighting, we compare the full model (hybrid kernel + feature weights) against the Static MKL baseline (hybrid kernel without feature weights), and quantify the specific improvement attributable to data-quality-driven weighting.

Statistical significance. Paired Wilcoxon signed-rank tests across the $5 \times 5 = 25$ fold results are used to check whether performance differences between the proposed method and each baseline are significant at the $p < 0.05$ level.

4.10 Implementation details

The framework is implemented in Python 3.10+ using the following stack:

- scikit-learn (v1.3+): SVM solver via SVC with precomputed kernel matrix. The precomputed kernel interface allows the custom adaptive hybrid kernel to be integrated without modifying the core optimization code.
- NumPy/SciPy: numerical computation, kernel matrix construction, and statistical tests.
- Pandas: data loading, preprocessing, and feature extraction.
- Matplotlib/Seaborn: visualization of results, noise degradation curves, and feature weight distributions.

Experiments run on a standard workstation (no GPU required; SVM training does not benefit from GPU acceleration at the dataset sizes we are considering). Code, experiment configurations, random seeds, and preprocessing parameters are documented for reproducibility, and the full codebase will be submitted alongside the thesis.

5. Timeline and Milestones

The timeline assumes a 16-week (4-month) Master’s project, with milestones aligned to the narrowed scope. The schedule leaves buffer for debugging and iteration, and writing begins early so it does not bottleneck the end of the project.

Phase	Weeks	Activities and deliverables
Phase 1: Data preparation & literature finalization	1–3	Acquire and preprocess OPPORTUNITY and CASAS datasets. Implement feature extraction pipelines. Generate synthetic noise variants for robustness testing. Finalize literature review. Deliverable: cleaned datasets, preprocessing code, noise injection module.
Phase 2: Core model implementation	4–7	Implement feature weight computation module (missingness + variance stability). Build the adaptive hybrid kernel function with precomputed kernel matrix. Integrate with scikit-learn SVM solver. Implement the hyperparameter grid search pipeline. Implement all three baselines. Deliverable: working model code, baseline implementations, initial results.
Phase 3: Experiments & evaluation	8–11	Run full cross-validation experiments on both datasets (5 seeds × 5 folds). Execute noise robustness experiments. Conduct ablation study. Perform statistical significance testing. Compile results tables and figures. Deliverable: complete experimental results, analysis.
Phase 4: Thesis writing & defense	12–16	Write methodology, results, and discussion chapters. Prepare visualizations and results tables. Address supervisor feedback on drafts. Prepare final manuscript and defense presentation. Deliverable: complete thesis document, defense slides.

Progress meetings with the thesis supervisor are scheduled biweekly. A brief progress report will be prepared for each meeting, documenting completed tasks, current challenges, and planned next steps. Writing begins in Phase 1 with the literature review and continues in parallel with implementation so it does not pile up at the end.

6. Expected Contributions

This thesis is expected to contribute the following to the field of machine learning for IoT healthcare data:

12. A unified adaptive hybrid kernel framework. The main methodological contribution is integrating data-quality-driven feature weighting directly into a composite linear-RBF kernel for SVM classification. Standard MKL learns static kernel weights; preprocessing-based feature weighting is decoupled from the kernel. The proposed framework instead modulates the kernel similarity computation itself using feature reliability indicators. As a result, the SVM's decision function naturally de-emphasizes unreliable features, which gives a principled mechanism for noise robustness in high-dimensional IoT data.
13. Empirical evidence on noise robustness. The thesis provides systematic empirical evidence on the effectiveness of the proposed approach through benchmarking on two IoT healthcare datasets. The noise injection experiments quantify the specific benefit of feature weighting for maintaining classifier performance under degraded sensor conditions, and give practical guidance for deploying SVM-based systems in noisy IoT environments.
14. Ablation analysis of component contributions. Comparing the full model against the static MKL baseline (hybrid kernel without feature weighting) isolates the contribution of the data-quality-driven feature weighting mechanism. The analysis clarifies when and how much feature weighting improves on the hybrid kernel alone, which in turn informs future research on the relative value of kernel composition versus feature weighting for different types of IoT data.
15. Reusable implementation. A documented, scikit-learn-compatible Python implementation will be provided so that other researchers can apply and extend the method to their own datasets. The implementation includes the feature weight computation module, the adaptive hybrid kernel function, and the complete experimental pipeline with configuration files for reproducibility.

7. Ethical Considerations

Several ethical dimensions bear on this work. The subsections below cover each one and the measures taken to handle it responsibly.

7.1 Data ethics and privacy

The research uses only publicly available, anonymized benchmark datasets (OPPORTUNITY and CASAS). Both have been released by their respective institutions for academic research and contain no personally identifiable information, so no additional ethical approval is required. No new data collection involving human participants is planned. Data handling follows privacy-by-design principles: secure local storage, no cloud uploads of raw data, and clear documentation of data provenance and usage terms.

7.2 Transparency and reproducibility

All experimental procedures, hyperparameter configurations, random seeds, and preprocessing steps are documented and submitted alongside the thesis. The codebase will be organized and commented so that results can be replicated independently. This follows standard practices for reproducible machine learning research and lets the claimed contributions be verified.

7.3 Limitations and responsible reporting

The thesis reports the limitations of the proposed method explicitly, including cases where it may not beat the baselines, its sensitivity to hyperparameter choices, and its assumptions about sensor noise. Results are reported with uncertainty quantification (standard deviations across repeated runs) and statistical significance testing. No clinical effectiveness claims are made, since the evaluation is on benchmark datasets in an offline research setting and not in a clinical deployment.

7.4 Broader impact considerations

While the method is evaluated in an academic setting, deploying it in real-world IoT healthcare systems would raise additional ethical considerations: patient consent for continuous monitoring, fairness across demographic groups with different device usage patterns, and regulatory requirements for clinical decision support systems. These matter but fall outside the scope of the current thesis. Any future clinical deployment would require independent ethical review, regulatory approval, and extensive clinical validation beyond what is done here.

8. Future Work

Several extensions of the proposed framework would be worth pursuing beyond this thesis:

Online/streaming adaptation. The current framework computes feature weights from the training data and applies them statically at inference. A natural extension is to update feature weights online as new data arrives, so the model can adapt to temporal changes in sensor reliability. This would require efficient incremental update rules for both the feature weights and the kernel mixture coefficient, while keeping the kernel matrix positive semi-definite.

Edge deployment optimization. For practical deployment on resource-constrained IoT edge devices, the kernel computation and prediction pipeline would need to be optimized for memory and latency. Candidates include random Fourier feature approximations of the RBF kernel, model compression through support vector pruning, and quantization of the kernel matrix.

Fairness and bias analysis. Whether the adaptive feature weighting introduces or mitigates demographic biases in predictions is an important question for healthcare applications. Future work could check performance consistency across patient subgroups defined by age, gender, or device type, and explore fairness-aware extensions of the weight optimization.

Joint optimization of feature weights and kernel parameters. The current design separates feature weight computation from SVM optimization for tractability. A more ambitious approach would optimize feature weights, kernel mixture coefficients, and SVM parameters jointly within a single framework, potentially yielding better solutions at the cost of non-convexity.

Additional kernel components. The current framework uses two kernel components (linear and RBF). Extending to three or more (for example adding a polynomial or sigmoid kernel) could capture more data structure, at the cost of a larger hyperparameter search space.

References

- Abdellatif, A. A. H., Singh, A., Aldribi, A., Ortega, A., & Ibrahim, M. (2022). A novel framework for fog-assisted smart healthcare system with workload optimization. *Computational Intelligence and Neuroscience*, 2022, 4174805. <https://doi.org/10.1155/2022/4174805>
- Al-amri, R., Murugesan, R. K., Man, M., Fareed, A., Al-Sharafi, M. A., & Alkahtani, A. A. (2021). A review of machine learning and deep learning techniques for anomaly detection in IoT data. *Applied Sciences*, 11(12), 5320. <https://doi.org/10.3390/app11125320>
- AL-Dhief, F. T., Latiff, N. M. A., Malik, N. N. N. A., et al. (2020). A survey of voice pathology surveillance systems based on internet of things and machine learning algorithms. *IEEE Access*, 8, 64514–64533. <https://doi.org/10.1109/access.2020.2984925>
- Al-hajjar, A. L. N., & Al-Qurabat, A. K. M. (2023). An overview of machine learning methods in enabling IoMT-based epileptic seizure detection. *The Journal of Supercomputing*. <https://doi.org/10.1007/s11227-023-05299-9>
- Cao, P., Liu, X., Yang, J., & Zhao, D. (2017). Sparse feature based kernel SVM for fault diagnosis. *Neural Computing and Applications*, 29(12), 1439–1449.
- Chavarriaga, R., Sagha, H., Calatroni, A., Digumarti, S. T., Tröster, G., Millán, J. R., & Roggen, D. (2013). The Opportunity challenge: A benchmark database for on-body sensor-based activity recognition. *Pattern Recognition Letters*, 34(15), 2033–2042. <https://doi.org/10.1016/j.patrec.2012.12.014>
- Cook, D., Crandall, A., Thomas, B., & Krishnan, N. (2013). CASAS: A smart home in a box. *IEEE Computer*, 46(7), 62–69. <https://doi.org/10.1109/MC.2012.328>
- Gönen, M., & Alpaydin, E. (2011). Multiple kernel learning algorithms. *Journal of Machine Learning Research*, 12, 2211–2268.
- Lin, C. F., & Wang, S. D. (2004). Training algorithms for fuzzy support vector machines with noisy data. *Pattern Recognition Letters*, 25(14), 1647–1656. <https://doi.org/10.1016/j.patrec.2004.06.009>
- Mahdavinejad, M. S., Rezvan, M., Barekatain, M., Adibi, P., Barnaghi, P., & Sheth, A. (2018). Machine learning for internet of things data analysis: A survey. *Digital Communications and Networks*, 4, 161–175. <https://doi.org/10.1016/j.dcan.2017.10.002>
- Muriira, L. M., Zhao, Z., & Min, G. (2018). Exploiting linear support vector machine for correlation-based high dimensional data classification in wireless sensor networks. *Sensors*, 18(9), 2840. <https://doi.org/10.3390/s18092840>
- Rakotomamonjy, A., Bach, F. R., Canu, S., & Grandvalet, Y. (2008). SimpleMKL. *Journal of Machine Learning Research*, 9, 2491–2521.
- Sonnenburg, S., Rätsch, G., Schäfer, C., & Schölkopf, B. (2006). Large scale multiple kernel learning. *Journal of Machine Learning Research*, 7, 1531–1565.

- Talaat, F. M., & El-Balka, R. M. (2023). Stress monitoring using wearable sensors: IoT techniques in medical field. *Neural Computing and Applications*, 35(25), 18571–18584.
<https://doi.org/10.1007/s00521-023-08681-z>
- Vallabhuni, S., & Debasis, K. (2025). Hybrid deep learning for IoT-based health monitoring with physiological event extraction. *Digital Health*, 11, 20552076251337848.
<https://doi.org/10.1177/20552076251337848>
- Wang, Y., Liu, L., & Wang, C. (2023). Trends in using deep learning algorithms in biomedical prediction systems. *Frontiers in Neuroscience*, 17, 1256351. <https://doi.org/10.3389/fnins.2023.1256351>
- Yang, X., Song, Q., & Cao, A. (2005). Weighted support vector machine for data classification. In *Proceedings of the IEEE International Joint Conference on Neural Networks (IJCNN)*, 2, 859–864.
<https://doi.org/10.1109/IJCNN.2005.1555965>
- Zhang, Q., Liu, D., Fan, Z., Lee, Y., & Li, Z. (2011). Feature and sample weighted support vector machine. In Y. Wang & T. Li (Eds.), *Knowledge Engineering and Management, Advances in Intelligent and Soft Computing*, 123. Springer. https://doi.org/10.1007/978-3-642-25661-5_47